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Enterprise VPN, Firewall, and Secure Access Implementation

Introduction

The project focuses on protecting enterprise communication and infrastructure through multiple security layers, including IPsec Site-to-Site VPN, Zone-Based Firewall, AAA authentication, SSH Version 2, OSPF authentication, device hardening, and centralized monitoring.

The simulated enterprise consists of a Headquarters network, Lahore branch, Karachi branch, and an ISP/WAN router providing inter-site connectivity. The implementation was validated using Cisco IOS verification commands, endpoint connectivity tests, VPN security associations, and administrative-access verification.

The project also included practical troubleshooting. In particular, initial LAN connectivity problems were investigated at the switching and GNS3 endpoint level before proceeding with VPN verification.

Assignment overview

The architecture was designed around the principle of **defense in depth**. Instead of relying on a single security mechanism, multiple independent controls were implemented.

Security Area	Technology Implemented	Purpose
VPN Security	IPsec / ISAKMP	Encrypt inter-site traffic
Firewall	Zone-Based Firewall	Control and inspect traffic
Authentication	AAA	Centralize administrative authentication
Remote Management	SSHv2	Secure device administration
Routing Security	OSPF MD5	Protect routing exchanges
Monitoring	Syslog	Centralize device events
Time Security	NTP	Synchronize timestamps
Network Monitoring	SNMPv3	Secure monitoring
Hardening	Cisco IOS security controls	Reduce attack surface

Objectives

The primary objectives of the project were:

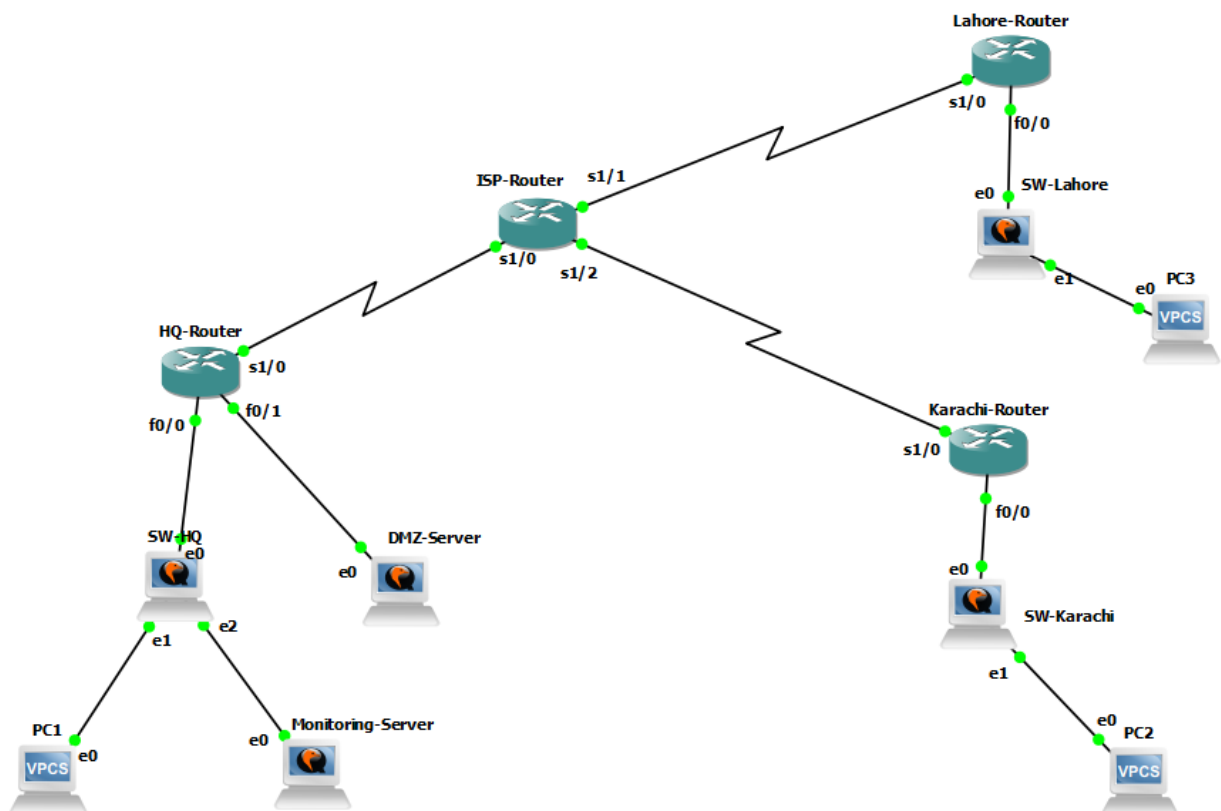
- Implement secure branch-to-branch communication using IPsec Site-to-Site VPN.
- Configure Zone-Based Firewall policies to protect internal networks.
- Deploy AAA authentication and SSH Version 2 for secure administration.
- Implement OSPF authentication and passive interfaces.
- Apply basic Cisco IOS device-hardening techniques.
- Implement Syslog, NTP, and SNMPv3 for monitoring and operational visibility.
- Verify all security controls using appropriate Cisco IOS commands.
- Troubleshoot connectivity and security issues encountered during implementation.

Tools and technologies

Tool / Technology	Role
GNS3	Network simulation platform
Cisco IOS 12.4	Router operating system
Cisco c3725	Enterprise router platform
VPCS	End-host simulation
OSPF	Dynamic routing
IPsec	VPN encryption
ISAKMP/IKE	VPN Phase 1 negotiation
Crypto Map	IPsec policy binding
ZBF	Stateful firewall
AAA	Authentication
SSHv2	Secure management
Syslog	Centralized logging
NTP	Time synchronization
SNMPv3	Secure monitoring

The Cisco c3725 platform was retained for the implementation because it provided the required IOS security functionality in the GNS3 environment.

Network topology



Ip addressing plan

The final implementation used the following addressing scheme.

Device	Interface	IP Address	Network / Purpose
HQ-Router	Fa0/0	192.168.10.1/24	HQ LAN
HQ-Router	Fa0/1	192.168.40.1/24	DMZ
HQ-Router	Serial1/0	10.0.1.2/30	HQ-ISP WAN
ISP-Router	Serial1/0	10.0.1.1/30	HQ WAN
ISP-Router	Serial1/1	10.0.2.1/30	Lahore WAN
ISP-Router	Serial1/2	10.0.3.1/30	Karachi WAN
Lahore-Router	Serial1/0	10.0.2.2/30	Lahore WAN
Lahore-Router	Fa0/0	192.168.20.1/24	Lahore LAN
Karachi-Router	Serial1/0	10.0.3.2/30	Karachi WAN
Karachi-Router	Fa0/0	192.168.30.1/24	Karachi LAN
PC1	e0	192.168.10.10/24	HQ Client
PC2	e0	192.168.30.10/24	Karachi Client
PC3	e0	192.168.20.10/24	Lahore Client
DMZ Server	e0	192.168.40.10/24	DMZ
Monitoring Server	e0	192.168.10.20/24	Monitoring

Gateway Configuration

Host	Default Gateway
PC1	192.168.10.1
PC2	192.168.30.1
PC3	192.168.20.1
DMZ Server	192.168.40.1
Monitoring Server	192.168.10.1

Security architecture

The network was designed using a layered security architecture.

Layer	Security Control	Security Objective
WAN Confidentiality	IPsec VPN	Encrypt inter-site traffic
Network Traffic Control	ZBF	Enforce stateful traffic policies
Administrative Security	AAA + SSH	Authenticate administrators
Routing Security	OSPF MD5	Prevent unauthorized routing updates
Monitoring	Syslog	Record security/network events
Time	NTP	Provide consistent timestamps
Management	SNMPv3	Secure network monitoring
Device Security	Hardening	Reduce attack surface

The reference report similarly uses layered controls covering VPN confidentiality, firewall traffic control, administrative access, routing integrity, and monitoring.

Implementation and configuration

HQ-Router:

```
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#hostname HQ-Router
HQ-Router(config)#interface FastEthernet0/0
HQ-Router(config-if)# description HQ-INSIDE-LAN
HQ-Router(config-if)# ip address 192.168.10.1 255.255.255.0
HQ-Router(config-if)#no shutdown
HQ-Router(config-if)#exit
*Mar  1 00:05:29.231: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:05:30.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
HQ-Router(config-if)#exit
HQ-Router(config)#interface FastEthernet0/1
HQ-Router(config-if)#description HQ-DMZ
HQ-Router(config-if)#ip address 192.168.40.1 255.255.255.0
HQ-Router(config-if)#no shutdown
HQ-Router(config-if)#exit
HQ-Router(config)#
*Mar  1 00:05:53.299: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:05:54.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
HQ-Router(config)#interface Serial1/0
HQ-Router(config-if)#description HQ-WAN-to-ISP
HQ-Router(config-if)#ip address 10.0.1.2 255.255.255.252
HQ-Router(config-if)# no shutdown
HQ-Router(config-if)#exit
HQ-Router(config)#end
```

Lahore-Router

```
Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#hostname Lahore-Router
Lahore-Router(config)#interface FastEthernet0/0
Lahore-Router(config-if)# description LAHORE-INSIDE-LAN
Lahore-Router(config-if)# ip address 192.168.20.1 255.255.255.0
Lahore-Router(config-if)#no shutdown
Lahore-Router(config-if)#exit
Lahore-Router(config)#
*Mar  1 00:07:18.595: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:07:19.595: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Lahore-Router(config)#interface Serial1/0
Lahore-Router(config-if)# description LAHORE-WAN-to-ISP
Lahore-Router(config-if)# ip address 10.0.2.2 255.255.255.252
Lahore-Router(config-if)# no shutdown
Lahore-Router(config-if)#exit
Lahore-Router(config)#end
```

Karachi-Router

```
Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#hostname Karachi-Router
Karachi-Router(config)#interface FastEthernet0/0
Karachi-Router(config-if)# description KARACHI-INSIDE-LAN
Karachi-Router(config-if)# ip address 192.168.30.1 255.255.255.0
Karachi-Router(config-if)# no shutdown
Karachi-Router(config-if)#exit
Karachi-Router(config)#
*Mar 1 00:07:45.987: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:07:46.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Karachi-Router(config)#interface Serial1/0
Karachi-Router(config-if)# description KARACHI-WAN-to-ISP
Karachi-Router(config-if)# ip address 10.0.3.2 255.255.255.252
Karachi-Router(config-if)# no shutdown
Karachi-Router(config-if)#exit
Karachi-Router(config)#end
```

ISP Router

```
ISP-Router#en
ISP-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#hostname ISP-Router
ISP-Router(config)#interface Serial1/0
ISP-Router(config-if)#description WAN-to-HQ
ISP-Router(config-if)# ip address 10.0.1.1 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)#exit
ISP-Router(config)#
*Mar 1 00:06:25.855: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
ISP-Router(config)#
*Mar 1 00:06:26.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
ISP-Router(config)#interface Serial1/1
ISP-Router(config-if)#description WAN-to-LAHORE
ISP-Router(config-if)# ip address 10.0.2.1 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)#exit
ISP-Router(config)#
*Mar 1 00:06:47.495: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
ISP-Router(config)#
*Mar 1 00:06:48.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
ISP-Router(config)#interface Serial1/2
ISP-Router(config-if)# description WAN-to-KARACHI
ISP-Router(config-if)# ip address 10.0.3.1 255.255.255.252
ISP-Router(config-if)# no shutdown
ISP-Router(config-if)#exit
ISP-Router(config)#
```

Test direct wan connectivity

Before OSPF, test each serial neighbor.

```
HQ-Router#ping 10.0.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/41/72 ms
HQ-Router#
```

```
Lahore-Router#ping 10.0.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/100/148 ms
Lahore-Router#
```

```
Karachi-Router#ping 10.0.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.3.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/108/136 ms
Karachi-Router#
```

Configure the vpcs

Now PC1.

```
PC1> ip 192.168.10.10/24 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.10.10/24
GATEWAY    : 192.168.10.1
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10003
RHOST:PORT : 127.0.0.1:10004
MTU        : 1500

PC1> 
```

```

PC3> ip 192.168.20.10/24 192.168.20.1
Checking for duplicate address...
PC1 : 192.168.20.10 255.255.255.0 gateway 192.168.20.1

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 192.168.20.10/24
GATEWAY    : 192.168.20.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10005
RHOST:PORT : 127.0.0.1:10006
MTU        : 1500

PC3> █

```

```

PC2> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10007
RHOST:PORT : 127.0.0.1:10008
MTU        : 1500

PC2> █

```

AAA authentication and SSH

AAA was implemented using the local Cisco IOS user database.

The administrative account was configured as:

Parameter	Value
Username	admin
Privilege	15
Authentication	Local database
SSH	Version 2
RSA Key	1024-bit
VTY Access	SSH only

AAA + SSH Secure Device Management on HQ-Router

```
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#username admin privilege 15 secret Admin@123
HQ-Router(config)#aaa new-model
HQ-Router(config)#aaa authentication login default local
HQ-Router(config)#service password-encryption
HQ-Router(config)#ip domain-name itsimplera.local
HQ-Router(config)#crypto key generate rsa modulus 1024
The name for the keys will be: HQ-Router.itsimplera.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

HQ-Router(config)#
*Mar  1 00:37:00.603: %SSH-5-ENABLED: SSH 1.99 has been enabled
HQ-Router(config)#ip ssh version 2
HQ-Router(config)#ip ssh time-out 60
HQ-Router(config)#ip ssh authentication-retries 3
HQ-Router(config)#$ #AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS#
HQ-Router(config)#line console 0
HQ-Router(config-line)#login authentication default
HQ-Router(config-line)#exec-timeout 10 0
HQ-Router(config-line)#logging synchronous
HQ-Router(config-line)#exit
HQ-Router(config)#line vty 0 4
HQ-Router(config-line)#login authentication default
HQ-Router(config-line)#transport input ssh
HQ-Router(config-line)#exec-timeout 10 0
HQ-Router(config-line)#logging synchronous
HQ-Router(config-line)#exit
HQ-Router(config)#end
```

Now Verify HQ

```
HQ-Router#
HQ-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
HQ-Router#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
HQ-Router#show running-config | include username
username admin privilege 15 secret 5 $1$w4iN$GCwStxgUYpEfxGV68owqN1
HQ-Router#show running-config | section line vty
line vty 0 4
  logging synchronous
  transport input ssh
HQ-Router#show crypto key mypubkey rsa
% Key pair was generated at: 00:37:00 UTC Mar 1 2002
Key name: HQ-Router.itsimplera.local
Storage Device: private-config
Usage: General Purpose Key
Key is not exportable.
Key Data:
 30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00D542C3
 EE330DB8 7A9D4DB3 CCA0045B 9ECCDF2A 356EF4C6 4C1DB465 CEBE7650 D64C34EA
 2F14B950 9751D80E 17769B60 D0808CA6 118B88CF 143EA017 E7824E2D 1A6C4CCC
 E095C699 8A1756DF C1F10F97 82933072 C4536B6F 9803011B C30F6DA2 F5942566
 04DC7079 B551A3D6 1597D95F 40DE0368 5A737C28 FDCCC764 52AC7054 75020301 0001
% Key pair was generated at: 00:37:02 UTC Mar 1 2002
Key name: HQ-Router.itsimplera.local.server
Temporary key
Usage: Encryption Key
Key is not exportable.
Key Data:
 307C300D 06092A86 4886F70D 01010105 00036B00 30680261 009BE1F7 11778A7A
 2B925837 8B19DF68 6A444528 E9746D04 8052E0FE B755F346 8783EF7E 50CB8764
 EF9EA2E7 95009F0C D56E331E E598F9D7 7AF9FB38 7DEAC1F6 4BFAFEAE 68F7F7D3
 41186C32 35EC118F 72FF8421 C7B26061 963B1D09 855927C2 2B020301 0001
HQ-Router#
```

AAA + SSH Secure Device Management on ISP-Router

```
ISP-Router#en
ISP-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP-Router(config)#username admin privilege 15 secret Admin@123
ISP-Router(config)#aaa new-model
ISP-Router(config)#aaa authentication login default local
ISP-Router(config)#service password-encryption
ISP-Router(config)#ip domain-name itsimplera.local
ISP-Router(config)#crypto key generate rsa modulus 1024
The name for the keys will be: ISP-Router.itsimplera.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ISP-Router(config)#
*Mar  1 00:39:00.299: %SSH-5-ENABLED: SSH 1.99 has been enabled
ISP-Router(config)#ip ssh version 2
ISP-Router(config)#ip ssh time-out 60
ISP-Router(config)#ip ssh authentication-retries 3
ISP-Router(config)#$ #AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS#
ISP-Router(config)#line console 0
ISP-Router(config-line)#login authentication default
ISP-Router(config-line)#exec-timeout 10 0
ISP-Router(config-line)#logging synchronous
ISP-Router(config-line)#exit
ISP-Router(config)#line vty 0 4
ISP-Router(config-line)#login authentication default
ISP-Router(config-line)#transport input ssh
ISP-Router(config-line)#exec-timeout 10 0
ISP-Router(config-line)#logging synchronous
ISP-Router(config-line)#exit
ISP-Router(config)#end
```

Verify

```
ISP-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
ISP-Router#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
ISP-Router#show running-config | section line vty
line vty 0 4
  logging synchronous
  transport input ssh
ISP-Router#
```

AAA + SSH Secure Device Management on Lahore-Router

```
Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#username admin privilege 15 secret Admin@123
Lahore-Router(config)#aaa new-model
Lahore-Router(config)#aaa authentication login default local
Lahore-Router(config)#service password-encryption
Lahore-Router(config)#ip domain-name itsimplera.local
Lahore-Router(config)#crypto key generate rsa modulus 1024
The name for the keys will be: Lahore-Router.itsimplera.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Lahore-Router(config)#
*Mar  1 00:40:17.267: %SSH-5-ENABLED: SSH 1.99 has been enabled
Lahore-Router(config)#ip ssh version 2
Lahore-Router(config)#ip ssh time-out 60
Lahore-Router(config)#ip ssh authentication-retries 3
Lahore-Router(config)#$ #AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS#
Lahore-Router(config)#line console 0
Lahore-Router(config-line)#login authentication default
Lahore-Router(config-line)#exec-timeout 10 0
Lahore-Router(config-line)#logging synchronous
Lahore-Router(config-line)#exit
Lahore-Router(config)#line vty 0 4
Lahore-Router(config-line)#login authentication default
Lahore-Router(config-line)#transport input ssh
Lahore-Router(config-line)#exec-timeout 10 0
Lahore-Router(config-line)#logging synchronous
Lahore-Router(config-line)#exit
Lahore-Router(config)#end
```

Verify

```
Lahore-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Lahore-Router#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
Lahore-Router#show running-config | section line vty
line vty 0 4
  logging synchronous
  transport input ssh
Lahore-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Lahore-Router#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
Lahore-Router#show running-config | section line vty
line vty 0 4
  logging synchronous
  transport input ssh
Lahore-Router#
```

AAA + SSH Secure Device Management on Karachi-Router

```
Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#username admin privilege 15 secret Admin@123
Karachi-Router(config)#aaa new-model
Karachi-Router(config)#aaa authentication login default local
Karachi-Router(config)#service password-encryption
Karachi-Router(config)#ip domain-name itsimplera.local
Karachi-Router(config)#crypto key generate rsa modulus 1024
The name for the keys will be: Karachi-Router.itsimplera.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Karachi-Router(config)#
*Mar  1 00:41:38.239: %SSH-5-ENABLED: SSH 1.99 has been enabled
Karachi-Router(config)#ip ssh version 2
Karachi-Router(config)#ip ssh time-out 60
Karachi-Router(config)#ip ssh authentication-retries 3
Karachi-Router(config)#$ #AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS#
Karachi-Router(config)#line console 0
Karachi-Router(config-line)#login authentication default
Karachi-Router(config-line)#exec-timeout 10 0
Karachi-Router(config-line)#logging synchronous
Karachi-Router(config-line)#exit
Karachi-Router(config)#line vty 0 4
Karachi-Router(config-line)#login authentication default
Karachi-Router(config-line)#transport input ssh
Karachi-Router(config-line)#exec-timeout 10 0
Karachi-Router(config-line)#logging synchronous
Karachi-Router(config-line)#exit
Karachi-Router(config)#end
```

Verify

```
Karachi-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Karachi-Router#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
Karachi-Router#show running-config | section line vty
line vty 0 4
  logging synchronous
  transport input ssh
Karachi-Router#
```

Testing real SSH

```
HQ-Router#
HQ-Router#ping 10.0.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/62/92 ms
HQ-Router#ssh -l admin 10.0.1.1

Password:
AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS
ISP-Router>en
% Error in authentication.

ISP-Router>exit

[Connection to 10.0.1.1 closed by foreign host]
HQ-Router#
```

OSPF base routing security

First we need complete connectivity between all LANs on HQ, ISP Lahore and Karachi Router.

```
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
HQ-Router(config)#router ospf 1
HQ-Router(config-router)# router-id 1.1.1.1
HQ-Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
HQ-Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
HQ-Router(config-router)# network 10.0.1.0 0.0.0.3 area 0
HQ-Router(config-router)#end
HQ-Router#
*Mar  1 00:18:43.523: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
```

```
ISP-Router#en
ISP-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP-Router(config)#router ospf 1
ISP-Router(config-router)# router-id 2.2.2.2
ISP-Router(config-router)# network 10.0.1.0 0.0.0.3 area 0
ISP-Router(config-router)#
*Mar  1 00:17:51.619: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial1/0 from LOADI
NG to FULL, Loading Done
ISP-Router(config-router)# network 10.0.2.0 0.0.0.3 area 0
ISP-Router(config-router)# network 10.0.3.0 0.0.0.3 area 0
ISP-Router(config-router)#end
ISP-Router#
*Mar  1 00:18:04.887: %SYS-5-CONFIG_I: Configured from console by console
ISP-Router#write memory
Building configuration...
[OK]
```

```
Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Lahore-Router(config)#router ospf 1
Lahore-Router(config-router)# router-id 3.3.3.3
Lahore-Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Lahore-Router(config-router)# network 10.0.2.0 0.0.0.3 area 0
Lahore-Router(config-router)#end
Lahore-Router#
*Mar  1 00:17:33.055: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial1/0 from LOADI
NG to FULL, Loading Done
Lahore-Router#write
*Mar  1 00:17:34.279: %SYS-5-CONFIG_I: Configured from console by console
Lahore-Router#write memory
Building configuration...
[OK]
```

```

Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#router ospf 1
Karachi-Router(config-router)# router-id 4.4.4.4
Karachi-Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Karachi-Router(config-router)# network 10.0.3.0 0.0.0.3 area 0
Karachi-Router(config-router)#end
Karachi-Router#
*Mar 1 00:17:17.595: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial1/0 from LOADI
NG to FULL, Loading Done
Karachi-Router#
*Mar 1 00:17:18.935: %SYS-5-CONFIG_I: Configured from console by console
Karachi-Router#write memory
Building configuration...
[OK]
Karachi-Router#

```

OSPF AUTHENTICATION

Use the same authentication key on all routers.

We'll use: OSPF key: OSPFSECURE

```

HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#router ospf 1
HQ-Router(config-router)# area 0 authentication message-digest
HQ-Router(config-router)#exit
HQ-Router(config)#interface Serial1/0
HQ-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
HQ-Router(config-if)#exit
HQ-Router(config)#end

```

```

ISP-Router#en
ISP-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#router ospf 1
ISP-Router(config-router)# area 0 authentication message-digest
ISP-Router(config-router)#exit
ISP-Router(config)#interface Serial1/0
ISP-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
ISP-Router(config-if)#exit
ISP-Router(config)#interface Serial1/1
ISP-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
ISP-Router(config-if)#
*Mar 1 00:24:19.499: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial1/0 from LOADI
NG to FULL, Loading Done
ISP-Router(config-if)#exit
ISP-Router(config)#interface Serial1/2
ISP-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
ISP-Router(config-if)#
*Mar 1 00:24:35.183: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial1/1 from FULL
to DOWN, Neighbor Down: Dead timer expired
ISP-Router(config-if)#exit
ISP-Router(config)#end

```

```
Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#router ospf 1
Lahore-Router(config-router)# area 0 authentication message-digest
Lahore-Router(config-router)#exit
Lahore-Router(config)#interface Serial1/0
Lahore-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
Lahore-Router(config-if)#exit
Lahore-Router(config)#end
Lahore-Router#
*Mar 1 00:24:52.607: %SYS-5-CONFIG_I: Configured from console by console
Lahore-Router# ip ospf message-digest-key 1 md5 OSPFSECURE
*Mar 1 00:24:57.123: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial1/0 from LOADI
NG to FULL, Loading Done
Lahore-Router#write memory
Building configuration...
[OK]
```

```
Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#router ospf 1
Karachi-Router(config-router)# area 0 authentication message-digest
Karachi-Router(config-router)#exit
Karachi-Router(config)#interface Serial1/0
Karachi-Router(config-if)# ip ospf message-digest-key 1 md5 OSPFSECURE
Karachi-Router(config-if)#exit
Karachi-Router(config)#end
Karachi-Router#
*Mar 1 00:25:00.555: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial1/0 from LOADI
NG to FULL, Loading Done
Karachi-Router#
*Mar 1 00:25:02.355: %SYS-5-CONFIG_I: Configured from console by console
Karachi-Router#write memory
Building configuration...
[OK]
```

configure Passive Interfaces

```
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#router ospf 1
HQ-Router(config-router)# passive-interface FastEthernet0/0
HQ-Router(config-router)# passive-interface FastEthernet0/1
HQ-Router(config-router)# no passive-interface Serial1/0
HQ-Router(config-router)#exit
HQ-Router(config)#end
HQ-Router#write
*Mar 1 00:32:55.779: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
HQ-Router#
```

```
Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#router ospf 1
Lahore-Router(config-router)# passive-interface FastEthernet0/0
Lahore-Router(config-router)# no passive-interface Serial1/0
Lahore-Router(config-router)#exit
Lahore-Router(config)#end
Lahore-Router#wri
*Mar 1 00:30:53.651: %SYS-5-CONFIG_I: Configured from console by console
Lahore-Router#write memory
Building configuration...
[OK]
Lahore-Router#
```

```
Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#router ospf 1
Karachi-Router(config-router)# passive-interface FastEthernet0/0
Karachi-Router(config-router)# no passive-interface Serial1/0
Karachi-Router(config-router)#exit
Karachi-Router(config)#en
% Ambiguous command: "en"
Karachi-Router(config)#end
Karachi-Router#write memory
*Mar 1 00:30:38.983: %SYS-5-CONFIG_I: Configured from console by console
Karachi-Router#write memory
Building configuration...
[OK]
Karachi-Router#
```

OSPF was deployed to provide dynamic routing between HQ, Lahore, Karachi, and the ISP transit network.

OSPF Process ID **1** and Area **0** were used.

The implementation included:

- OSPF dynamic routing
- MD5 authentication
- Passive interfaces
- Secure routing adjacency

HQ-Router was successfully observed forming an OSPF adjacency with the ISP-Router.

The verified neighbor included: 2.2.2.2 and reached: FULL

Assign IPs to Monitoring-Server and DMZ-Server

Monitoring-Server:

```
localhost:~# ip addr add 192.168.10.20/24 dev eth0
localhost:~# ip link set eth0 up
localhost:~# ip route add default via 192.168.10.1
localhost:~# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 0c:87:43:d6:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.20/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::e87:43ff:fed6:0/64 scope link
        valid_lft forever preferred_lft forever
localhost:~# ping -c 5 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: seq=0 ttl=255 time=112.269 ms
64 bytes from 192.168.10.1: seq=1 ttl=255 time=42.525 ms
64 bytes from 192.168.10.1: seq=2 ttl=255 time=39.175 ms
64 bytes from 192.168.10.1: seq=3 ttl=255 time=19.114 ms
64 bytes from 192.168.10.1: seq=4 ttl=255 time=24.525 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 19.114/47.521/112.269 ms
localhost:~# cat > /etc/network/interfaces << 'EOF'
> auto lo
> iface lo inet loopback
>
> auto eth0
> iface eth0 inet static
> address 192.168.10.20
> netmask 255.255.255.0
> gateway 192.168.10.1
> EOF
```

DMZ-Server

```
localhost:~# ls
localhost:~# ip addr add 192.168.40.10/24 dev eth0
localhost:~# ip link set eth0 up
localhost:~# ip route add default via 192.168.40.1
localhost:~# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 0c:f2:90:39:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.40.10/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::ef2:90ff:fe39:0/64 scope link tentative
        valid_lft forever preferred_lft forever
localhost:~# ping -c 5 192.168.40.1
PING 192.168.40.1 (192.168.40.1): 56 data bytes
64 bytes from 192.168.40.1: seq=0 ttl=255 time=73.025 ms
64 bytes from 192.168.40.1: seq=1 ttl=255 time=12.293 ms
64 bytes from 192.168.40.1: seq=2 ttl=255 time=23.165 ms
64 bytes from 192.168.40.1: seq=3 ttl=255 time=34.472 ms
64 bytes from 192.168.40.1: seq=4 ttl=255 time=51.932 ms

--- 192.168.40.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 12.293/38.977/73.025 ms
localhost:~# cat > /etc/network/interfaces << 'EOF'
> auto lo
> iface lo inet loopback
>
> auto eth0
> iface eth0 inet static
> address 192.168.40.10
> netmask 255.255.255.0
> gateway 192.168.40.1
> EOF
localhost:~# hostname DMZ-Server
DMZ-Server:~# echo "DMZ-Server" > /etc/hostname
DMZ-Server:~#
```

Set up Syslog, NTP, and SNMP — the actual monitoring stack

We'll use Monitoring-Server (192.168.10.20) as the central syslog/NTP/SNMP receiver.

First, on Monitoring-Server, install and start the required services (Alpine uses `apk`):

```
localhost:~# hostname Monitoring-Server
Monitoring-Server:~# echo "Monitoring-Server" > /etc/hostname
Monitoring-Server:~# apk update
3.24.1 [/media/cdrom/apks]
3.24.1 [/media/sda1/apks]
OK: 95 distinct packages available
Monitoring-Server:~# apk add rsyslog chrony net-snmp
ERROR: unable to select packages:
  net-snmp (no such package):
    required by: world[net-snmp]
  rsyslog (no such package):
    required by: world[rsyslog]
Monitoring-Server:~# cat >> /etc/rsyslog.conf << 'EOF'
> module(load="imudp")
> input(type="imudp" port="514")
> EOF
Monitoring-Server:~# rc-service rsyslog start
* rc-service: service `rsyslog' does not exist
Monitoring-Server:~# rc-update add rsyslog default
* rc-update: service `rsyslog' does not exist
Monitoring-Server:~# cat > /etc/chrony/chrony.conf << 'EOF'
> local stratum 10
> allow 192.168.0.0/16
> allow 10.0.0.0/8
> EOF
-sh: can't create /etc/chrony/chrony.conf: nonexistent directory
Monitoring-Server:~# rc-service chronyd start
* rc-service: service `chronyd' does not exist
Monitoring-Server:~# rc-update add chronyd default
* rc-update: service `chronyd' does not exist
Monitoring-Server:~# rc-service snmpd start
* rc-service: service `snmpd' does not exist
Monitoring-Server:~# rc-update add snmpd default
* rc-update: service `snmpd' does not exist
Monitoring-Server:~#
```

Configure Syslog + NTP + SNMPv3 on all 4 routers

```
HQ-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#logging host 192.168.10.20
HQ-Router(config)#logging trap informational
HQ-Router(config)#logging source-interface FastEthernet0/0
HQ-Router(config)#ntp server 192.168.10.20
HQ-Router(config)#snmp-server group SNMPGROUP v3 priv
HQ-Router(config)#$ROUP v3 auth sha SnmpAuth123 priv aes 128 SnmpPriv123
HQ-Router(config)#end
```

```
Karachi-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#logging host 192.168.10.20
Karachi-Router(config)#logging trap informational
Karachi-Router(config)#logging source-interface FastEthernet0/0
Karachi-Router(config)#ntp server 192.168.10.20
Karachi-Router(config)#snmp-server group SNMPGROUP v3 priv
Karachi-Router(config)#$ROUP v3 auth sha SnmpAuth123 priv aes 128 SnmpPriv123
Karachi-Router(config)#end
```

```
Lahore-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#logging host 192.168.10.20
Lahore-Router(config)#logging trap informational
Lahore-Router(config)#logging source-interface FastEthernet0/0
Lahore-Router(config)#ntp server 192.168.10.20
Lahore-Router(config)#snmp-server group SNMPGROUP v3 priv
Lahore-Router(config)#$ROUP v3 auth sha SnmpAuth123 priv aes 128 SnmpPriv123
Lahore-Router(config)#end
```

```
ISP-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#logging host 192.168.10.20
ISP-Router(config)#logging trap informational
ISP-Router(config)#logging source-interface FastEthernet0/0
ISP-Router(config)#ntp server 192.168.10.20
ISP-Router(config)#snmp-server group SNMPGROUP v3 priv
ISP-Router(config)#$ROUP v3 auth sha SnmpAuth123 priv aes 128 SnmpPriv123
ISP-Router(config)#end
```

Route Filtering with Prefix-Lists

```
ISP-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#ip prefix-list HQ-ONLY seq 5 permit 192.168.10.0/24
ISP-Router(config)#ip prefix-list HQ-ONLY seq 10 deny 0.0.0.0/0 le 32
ISP-Router(config)#router ospf 1
ISP-Router(config-router)#distribute-list prefix HQ-ONLY out Serial1/1
% Interface not allowed with OUT in case of OSPF
ISP-Router(config-router)#end
ISP-Router#write memory
Building configuration...

*Mar 1 01:01:26.415: %SYS-5-CONFIG_I: Configured from console by admin on console[OK]
ISP-Router#
```

SSH login + enable-mode escalation with Enable@123

```
HQ-Router#ssh -l admin 10.0.2.2

Password:
AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS
Lahore-Router>ls
Translating "ls"

Translating "ls"

% Unknown command or computer name, or unable to find computer address
Lahore-Router>
Lahore-Router>en
Password:
Lahore-Router#
Lahore-Router#
```

PFS on IPsec

```
HQ-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#crypto map HQ-LAHORE-VPN 10 ipsec-isakmp
HQ-Router(config-crypto-map)#set pfs group2
HQ-Router(config-crypto-map)#crypto map HQ-LAHORE-VPN 20 ipsec-isakmp
HQ-Router(config-crypto-map)#set pfs group2
HQ-Router(config-crypto-map)#end
```

```
Karachi-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#crypto map HQ-LAHORE-VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Karachi-Router(config-crypto-map)#set pfs group2
Karachi-Router(config-crypto-map)#crypto map HQ-LAHORE-VPN 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Karachi-Router(config-crypto-map)#set pfs group2
Karachi-Router(config-crypto-map)#end
```

```
Lahore-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#crypto map HQ-LAHORE-VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Lahore-Router(config-crypto-map)#set pfs group2
Lahore-Router(config-crypto-map)#crypto map HQ-LAHORE-VPN 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Lahore-Router(config-crypto-map)#set pfs group2
Lahore-Router(config-crypto-map)#end
```

```
ISP-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#crypto map HQ-LAHORE-VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
ISP-Router(config-crypto-map)#set pfs group2
ISP-Router(config-crypto-map)#crypto map HQ-LAHORE-VPN 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
ISP-Router(config-crypto-map)#set pfs group2
ISP-Router(config-crypto-map)#end
```

IPSEC SITE-TO-SITE VPN

IPsec VPN was implemented to protect communication between the HQ and branch networks over the simulated ISP/WAN infrastructure.

The VPN implementation consisted of:

- IKE/ISAKMP Phase 1
- Pre-shared authentication
- IPsec Phase 2
- Transform set
- Interesting-traffic ACL
- Crypto map
- Crypto map applied to the WAN interface

IKE PHASE 1

```
HQ-Router#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite
  encryption algorithm: DES - Data Encryption Standard (56 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #1 (768 bit)
  lifetime:            86400 seconds, no volume limit
HQ-Router#show crypto isakmp key
Keyring      Hostname/Address      Preshared Key
default      10.0.2.2                    VPN@123
              10.0.3.2                    VPN@456
```

IPSEC PHASE 2

The transform set used:

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac mode tunnel
```

The VPN interesting traffic was defined using extended ACLs.

```
HQ-Router#show crypto ipsec transform-set
Transform set VPN-SET: { esp-aes esp-sha-hmac }
will negotiate = { Tunnel, },
```

CRYPTO MAP

```
HQ-Router#show crypto map
Crypto Map "HQ-LAHORE-VPN" 10 ipsec-isakmp
  Peer = 10.0.2.2
  Extended IP access list VPN-HQ-LAHORE
    access-list VPN-HQ-LAHORE permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  Current peer: 10.0.2.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): Y
  DH group: group2
  Transform sets={
    VPN-SET,
  }

Crypto Map "HQ-LAHORE-VPN" 20 ipsec-isakmp
  Peer = 10.0.3.2
  Extended IP access list VPN-HQ-KARACHI
    access-list VPN-HQ-KARACHI permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
  Current peer: 10.0.3.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): Y
  DH group: group2
  Transform sets={
    VPN-SET,
  }
  Interfaces using crypto map HQ-LAHORE-VPN:
    Serial1/0
```

VPN verification

The VPN was successfully established after generating interesting traffic.

The following verification output demonstrated an active IKE security association: `show crypto isakmp sa`

Result:

```
HQ-Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.2.2     10.0.1.2     QM_IDLE        1001    0  ACTIVE
10.0.3.2     10.0.1.2     QM_IDLE        1002    0  ACTIVE

IPv6 Crypto ISAKMP SA
```

QM_IDLE confirms that IKE Phase 1 negotiation completed successfully.

ZONE-BASED Firewall Configuration

Zone-Based Firewall provides stateful traffic inspection by grouping interfaces into security zones.

The planned security model consists of:

Zone	Purpose
INSIDE	Trusted enterprise LAN
DMZ	Public/service resources
OUTSIDE	Untrusted WAN

The HQ router contains:

- HQ LAN: 192.168.10.0/24
- DMZ: 192.168.40.0/24
- WAN: 10.0.1.0/30

The firewall architecture was designed to control traffic between trusted and untrusted interfaces while allowing legitimate sessions.

Create the security zones

```
HQ-Router#en
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#zone security INSIDE
HQ-Router(config-sec-zone)#zone security OUTSIDE
HQ-Router(config-sec-zone)#zone security DMZ
HQ-Router(config-sec-zone)#end
HQ-Router#
*Mar 1 00:54:01.643: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#show zone security
zone self
  Description: System defined zone

zone INSIDE

zone OUTSIDE

zone DMZ

HQ-Router#
```

HQ-LAN

```
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#interface FastEthernet0/0
HQ-Router(config-if)# zone-member security INSIDE
HQ-Router(config-if)#exit
HQ-Router(config)#
```

HQ DMZ

```
HQ-Router(config)#interface FastEthernet0/1
HQ-Router(config-if)# zone-member security DMZ
HQ-Router(config-if)#exit
```

WAN

```
HQ-Router(config)#interface Serial1/0
HQ-Router(config-if)# zone-member security OUTSIDE
HQ-Router(config-if)#exit
HQ-Router(config)#END
HQ-Router#
*Mar  1 02:04:27.763: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
```

Verify

```
HQ-Router#show zone security
zone self
  Description: System defined zone

zone INSIDE
  Member Interfaces:
    FastEthernet0/0

zone OUTSIDE
  Member Interfaces:
    Serial1/0

zone DMZ
  Member Interfaces:
    FastEthernet0/1

HQ-Router#
```

Creating the INSIDE and OUTSIDE class map

```
HQ-Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
HQ-Router(config)#class-map type inspect match-any INSIDE-OUTSIDE-TRAFFIC
HQ-Router(config-cmap)# match protocol icmp
HQ-Router(config-cmap)# match protocol tcp
HQ-Router(config-cmap)# match protocol udp
HQ-Router(config-cmap)#exit
HQ-Router(config)#
```

Create the policy map

```
HQ-Router(config)#policy-map type inspect INSIDE-TO-OUTSIDE-POLICY
HQ-Router(config-pmap)# class type inspect INSIDE-OUTSIDE-TRAFFIC
HQ-Router(config-pmap-c)# inspect
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)# class class-default
HQ-Router(config-pmap-c)#drop
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)#exit
HQ-Router(config)#
```

Create the zone pair

```
HQ-Router(config)#$SIDE-TO-OUTSIDE source INSIDE destination OUTSIDE
HQ-Router(config-sec-zone-pair)#$licy type inspect INSIDE-TO-OUTSIDE-POLICY
HQ-Router(config-sec-zone-pair)#
```

INSIDE → DMZ policy

```
HQ-Router(config-sec-zone-pair)#$type inspect match-any INSIDE-DMZ-TRAFFIC
HQ-Router(config-cmap)# match protocol icmp
HQ-Router(config-cmap)# match protocol tcp
HQ-Router(config-cmap)# match protocol udp
HQ-Router(config-cmap)#exit
HQ-Router(config)#
```

Policy

```
HQ-Router(config)#policy-map type inspect INSIDE-TO-DMZ-POLICY
HQ-Router(config-pmap)# class type inspect INSIDE-DMZ-TRAFFIC
HQ-Router(config-pmap-c)# inspect
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)# class class-default
HQ-Router(config-pmap-c)#drop
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)#exit
HQ-Router(config)#
```

Zone pair:

```
HQ-Router(config)#$security INSIDE-TO-DMZ source INSIDE destination DMZ
HQ-Router(config-sec-zone-pair)#$licy type inspect INSIDE-TO-DMZ-POLICY
HQ-Router(config-sec-zone-pair)#exit
HQ-Router(config)#
```

DMZ → OUTSIDE policy

```
HQ-Router(config)#class-map type inspect match-any DMZ-OUTSIDE-TRAFFIC
HQ-Router(config-cmap)# match protocol icmp
HQ-Router(config-cmap)# match protocol tcp
HQ-Router(config-cmap)# match protocol udp
HQ-Router(config-cmap)#exit
HQ-Router(config)#policy-map type inspect DMZ-TO-OUTSIDE-POLICY
HQ-Router(config-pmap)# class type inspect DMZ-OUTSIDE-TRAFFIC
HQ-Router(config-pmap-c)# inspect
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)# class class-default
HQ-Router(config-pmap-c)#drop
HQ-Router(config-pmap-c)#exit
HQ-Router(config-pmap)#exit
HQ-Router(config)#$security DMZ-TO-OUTSIDE source DMZ destination OUTSIDE
HQ-Router(config-sec-zone-pair)#$licy type inspect DMZ-TO-OUTSIDE-POLICY
HQ-Router(config-sec-zone-pair)#exit
HQ-Router(config)#end
HQ-Router#
*Mar 1 02:11:29.915: %SYS-5-CONFIG_I: Configured from console by console
HQ-Router#write memory
Building configuration...
[OK]
HQ-Router#
```

Verify the ZBF configuration

```
HQ-Router#show zone-pair security
Zone-pair name INSIDE-TO-OUTSIDE
  Source-Zone INSIDE Destination-Zone OUTSIDE
  service-policy INSIDE-TO-OUTSIDE-POLICY
Zone-pair name INSIDE-TO-DMZ
  Source-Zone INSIDE Destination-Zone DMZ
  service-policy INSIDE-TO-DMZ-POLICY
Zone-pair name DMZ-TO-OUTSIDE
  Source-Zone DMZ Destination-Zone OUTSIDE
  service-policy DMZ-TO-OUTSIDE-POLICY
```

Verification from PC1 for VPN tunneling

```
PC1> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=40.863 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=36.693 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=21.697 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=41.601 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=49.031 ms

PC1> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=254 time=115.467 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=254 time=111.895 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=254 time=141.069 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=254 time=133.928 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=254 time=153.267 ms

PC1> ping 192.168.20.10
84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=166.645 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=134.692 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=152.247 ms
^C
PC1> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=253 time=123.317 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=253 time=115.974 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=253 time=118.399 ms
^C
PC1> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=61 time=209.130 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=61 time=141.334 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=61 time=164.337 ms
^C
PC1>
```

HQ↔Karachi IPsec

```
HQ-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-Router(config)#crypto isakmp key VPN@456 address 10.0.3.2
HQ-Router(config)#ip access-list extended VPN-HQ-KARACHI
HQ-Router(config-ext-nacl)#92.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
HQ-Router(config-ext-nacl)#exit
HQ-Router(config)#crypto map HQ-LAHORE-VPN 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
HQ-Router(config-crypto-map)#set peer 10.0.3.2
HQ-Router(config-crypto-map)#set transform-set VPN-SET
HQ-Router(config-crypto-map)#match address VPN-HQ-KARACHI
HQ-Router(config-crypto-map)#exit
HQ-Router(config)#end
```

```

Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#crypto isakmp policy 10
Karachi-Router(config-isakmp)#encr aes
Karachi-Router(config-isakmp)#authentication pre-share
Karachi-Router(config-isakmp)# group 2
Karachi-Router(config-isakmp)#exit
Karachi-Router(config)#crypto isakmp key VPN@456 address 10.0.1.2
Karachi-Router(config)#%c transform-set VPN-SET esp-aes esp-sha-hmac
Karachi-Router(cfg-crypto-trans)#ip access-list extended VPN-KARACHI-HQ
Karachi-Router(config-ext-nacl)#92.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
Karachi-Router(config-ext-nacl)#exit
Karachi-Router(config)#crypto map KARACHI-HQ-VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Karachi-Router(config-crypto-map)#set peer 10.0.1.2
Karachi-Router(config-crypto-map)#set transform-set VPN-SET
Karachi-Router(config-crypto-map)#match address VPN-KARACHI-HQ
Karachi-Router(config-crypto-map)#exit
Karachi-Router(config)#interface Serial1/0
Karachi-Router(config-if)#crypto map KARACHI-HQ-VPN
Karachi-Router(config-if)#e
*Mar 1 00:36:58.351: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Karachi-Router(config-if)#exit
Karachi-Router(config)#end

```

Zone-Based Firewall on Lahore-Router

```

Lahore-Router#en
Lahore-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Lahore-Router(config)#zone security INSIDE
Lahore-Router(config-sec-zone)#zone security OUTSIDE
Lahore-Router(config-sec-zone)#exit
Lahore-Router(config)#class-map type inspect match-any LAHORE-INSIDE-OUTSIDE
Lahore-Router(config-cmap)#match protocol icmp
Lahore-Router(config-cmap)#match protocol tcp
Lahore-Router(config-cmap)#match protocol udp
Lahore-Router(config-cmap)#exit
Lahore-Router(config)#policy-map type inspect LAHORE-INSIDE-TO-OUTSIDE-POLICY
Lahore-Router(config-pmap)#class type inspect LAHORE-INSIDE-OUTSIDE
Lahore-Router(config-pmap-c)#inspect
Lahore-Router(config-pmap-c)#class class-default
Lahore-Router(config-pmap-c)#drop
Lahore-Router(config-pmap-c)#exit
Lahore-Router(config-pmap)#$SIDE-TO-OUTSIDE source INSIDE destination OUTSIDE
Lahore-Router(config-sec-zone-pair)#$nspect LAHORE-INSIDE-TO-OUTSIDE-POLICY
Lahore-Router(config-sec-zone-pair)#exit
Lahore-Router(config)#interface FastEthernet0/0
Lahore-Router(config-if)#zone-member security INSIDE
Lahore-Router(config-if)#exit
Lahore-Router(config)#interface Serial1/0
Lahore-Router(config-if)#zone-member security OUTSIDE
Lahore-Router(config-if)#exit
Lahore-Router(config)#end

```

Zone-Based Firewall on Karachi-Router

```
Karachi-Router#en
Karachi-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Karachi-Router(config)#zone security INSIDE
Karachi-Router(config-sec-zone)#zone security OUTSIDE
Karachi-Router(config-sec-zone)#exit
Karachi-Router(config)#$type inspect match-any KARACHI-INSIDE-OUTSIDE
Karachi-Router(config-cmap)#match protocol icmp
Karachi-Router(config-cmap)#match protocol tcp
Karachi-Router(config-cmap)#match protocol udp
Karachi-Router(config-cmap)#exit
Karachi-Router(config)#$type inspect KARACHI-INSIDE-TO-OUTSIDE-POLICY
Karachi-Router(config-pmap)#class type inspect KARACHI-INSIDE-OUTSIDE
Karachi-Router(config-pmap-c)#inspect
Karachi-Router(config-pmap-c)#class class-default
Karachi-Router(config-pmap-c)#drop
Karachi-Router(config-pmap-c)#exit
Karachi-Router(config-pmap)#$TSIDE source INSIDE destination OUTSIDE
Karachi-Router(config-sec-zone-pair)#$inspect KARACHI-INSIDE-TO-OUTSIDE-POLICY
Karachi-Router(config-sec-zone-pair)#exit
Karachi-Router(config)#interface FastEthernet0/0
Karachi-Router(config-if)#zone-member security INSIDE
Karachi-Router(config-if)#exit
Karachi-Router(config)#interface Serial1/0
Karachi-Router(config-if)#zone-member security OUTSIDE
Karachi-Router(config-if)#exit
Karachi-Router(config)#end
```

SYSLOG, NTP AND SNMPv3

Centralized monitoring was implemented to improve visibility into network events.

Syslog

Syslog provides centralized collection of device-generated messages. The routers were configured to forward relevant events to the monitoring server.

```
HQ-Router#show running-config | include logging
logging source-interface FastEthernet0/0
logging 192.168.10.20
logging synchronous
logging synchronous
logging synchronous
HQ-Router#show logging
Syslog logging: enabled (12 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 53 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 0 messages logged, xml disabled,
filtering disabled
Buffer logging: disabled, xml disabled,
filtering disabled
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped

Trap logging: level informational, 57 message lines logged
Logging to 192.168.10.20 (udp port 514, audit disabled,
authentication disabled, encryption disabled, link up),
5 message lines logged,
0 message lines rate-limited,
0 message lines dropped-by-MD,
xml disabled, sequence number disabled
filtering disabled
```

NTP

NTP was configured to provide consistent time synchronization across network devices.

Accurate time is important for:

- troubleshooting
- event correlation
- security investigations
- Syslog timestamps

On HQ-Router

```
HQ-Router#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1900)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec
HQ-Router#show ntp associations

      address      ref clock      st when poll reach delay offset  disp
~192.168.10.20    0.0.0.0        16   -   64    0    0.0   0.00 16000.
* master (syncd), # master (unsyncd), + selected, - candidate, ~ configured
HQ-Router#
```

On Lahore-Router

```
Lahore-Router#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1900)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec
Lahore-Router#show ntp associations

      address      ref clock      st when poll reach delay offset  disp
~192.168.10.20    0.0.0.0        16   -   64    0    0.0   0.00 16000.
* master (syncd), # master (unsyncd), + selected, - candidate, ~ configured
Lahore-Router#
```

On Karachi-Router

```
Karachi-Router#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1900)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec
Karachi-Router#show ntp associations

      address      ref clock      st when poll reach delay offset  disp
~192.168.10.20    0.0.0.0        16   -   64    0    0.0   0.00 16000.
* master (syncd), # master (unsyncd), + selected, - candidate, ~ configured
Karachi-Router#
```

SNMPv3

SNMPv3 was selected because it provides authenticated and privacy-protected management traffic.

The implementation uses:

- SNMPv3 user
- authentication
- privacy/encryption
- secure management access

On HQ-Router:

```
HQ-Router#show snmp user
User name: SNMPUSER
Engine ID: 800000090300C20410DC0000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: SNMPGROUP

HQ-Router#show snmp group
groupname: ILMI                security model:v1
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                security model:v2c
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: SNMPGROUP           security model:v3 priv
readview : v1default           writeview: <no writeview specified>
notifyview: <no notifyview specified>
row status: active
```

On Lahore-Router:

```
Lahore-Router#show snmp user
User name: SNMPUSER
Engine ID: 800000090300C2020A2C0000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: SNMPGROUP

Lahore-Router#show snmp group
groupname: ILMI                security model:v1
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                security model:v2c
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: SNMPGROUP           security model:v3 priv
readview : v1default           writeview: <no writeview specified>
notifyview: <no notifyview specified>
row status: active
```

On Karachi-Router:

```
Karachi-Router#show snmp user
User name: SNMPUSER
Engine ID: 800000090300C2030A490000
storage-type: nonvolatile      active
Authentication Protocol: SHA
Privacy Protocol: AES128
Group-name: SNMPPGROUP

Karachi-Router#show snmp group
groupname: ILMI                security model:v1
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                security model:v2c
readview : *ilmi               writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: SNMPPGROUP          security model:v3 priv
readview : v1default           writeview: <no writeview specified>
notifyview: <no notifyview specified>
row status: active
```

Device Hardening

Basic device hardening was applied to reduce the attack surface.

Implemented controls included:

- AAA authentication
- SSH-only VTY access
- Telnet disabled
- Password encryption
- Login/MOTD banner
- Console timeout
- VTY timeout
- Unused interface shutdown
- Unnecessary services disabled
- Secure administrative access

The login banner used in the environment was: AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS

Verification

```
HQ-Router#show running-config
Building configuration...

Current configuration : 4312 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname HQ-Router
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$F76P$EEcDeVM6Zjj1xcqju060b.
!
aaa new-model
!
!
aaa authentication login default local
!
!
aaa session-id common
memory-size iomem 5
HQ-Router#show running-config | include service
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
  service-policy type inspect INSIDE-TO-OUTSIDE-POLICY
  service-policy type inspect INSIDE-TO-DMZ-POLICY
  service-policy type inspect DMZ-TO-OUTSIDE-POLICY
HQ-Router#show running-config | section line
line con 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  logging synchronous
  transport input ssh
```

Verification and Testing

All major security controls were verified using Cisco IOS commands and end-to-end connectivity tests.

Security Feature	Verification Command	Expected Result	Status
LAN connectivity	ping gateway	Successful replies	Done
WAN connectivity	ping WAN peer	Successful replies	Done
OSPF	show ip ospf neighbor	FULL adjacency	Done
OSPF authentication	show ip ospf interface	MD5 enabled	Done
AAA	Login test	admin authenticated	Done
SSH	show ip ssh	Version 2.0	Done
IPsec Phase 1	show crypto isakmp sa	QM_IDLE	Done
IPsec Phase 2	show crypto ipsec sa	ACTIVE SAs	Done
VPN traffic	ping 192.168.20.10	Successful	Done
ZBF	show policy-map type inspect	Counters active	Done
Syslog	show logging	Logs generated	Done
NTP	show ntp status	Synchronization	Done
SNMPv3	show snmp user	User present	Done

Local LAN Connectivity

After replacing/reconnecting the Lahore switch and running the VPCS endpoints inside the GNS3 VM, local connectivity was successfully restored.

HQ

```
HQ-Router#ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
HQ-Router#
```

Lahore

```
Lahore-Router#ping 192.168.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Lahore-Router#
```

Karachi

```
Karachi-Router#ping 192.168.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/8 ms
Karachi-Router#
```

VPN END-TO-END Connectivity

```
PC1> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=19.658 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=26.971 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=16.319 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=46.304 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=40.869 ms

PC1> ping 192.168.20.10

192.168.20.10 icmp_seq=1 timeout
192.168.20.10 icmp_seq=2 timeout
192.168.20.10 icmp_seq=3 timeout
192.168.20.10 icmp_seq=4 timeout
192.168.20.10 icmp_seq=5 timeout
^C
PC1> ping 192.168.30.10

192.168.30.10 icmp_seq=1 timeout
84 bytes from 192.168.30.10 icmp_seq=2 ttl=61 time=131.476 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=61 time=122.746 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=61 time=143.298 ms
^C
PC1>
PC1> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=61 time=203.656 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=61 time=116.743 ms
^C
PC1> ping 192.168.20.10

192.168.20.10 icmp_seq=1 timeout
192.168.20.10 icmp_seq=2 timeout
^C
PC1> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=40.863 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=36.693 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=21.697 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=41.601 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=49.031 ms

PC1> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=254 time=115.467 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=254 time=111.895 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=254 time=141.069 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=254 time=133.928 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=254 time=153.267 ms

PC1> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=166.645 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=134.692 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=152.247 ms
^C
PC1> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=253 time=123.317 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=253 time=115.974 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=253 time=118.399 ms
^C
PC1> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=61 time=209.130 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=61 time=141.334 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=61 time=164.337 ms
^C
PC1>
PC1>
PC1> ping 192.168.30.10

192.168.30.10 icmp_seq=1 timeout
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=262.815 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=124.737 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=123.639 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=152.883 ms
```

IPSEC Security Association

The following output provided direct evidence of VPN operation: state QM_IDLE and:

```
#pkts encaps: 14
#pkts encrypt: 14
#pkts decaps: 4
#pkts decrypt: 4
```

The active Security Association showed: transform: esp-aes esp-sha-hmac, Status: ACTIVE

This confirms that actual packets traversed the IPsec tunnel.

```
HQ-Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.2.2     10.0.1.2     QM_IDLE       1001      0 ACTIVE
10.0.3.2     10.0.1.2     QM_IDLE       1002      0 ACTIVE

IPv6 Crypto ISAKMP SA

HQ-Router#show crypto ipsec sa
interface: Serial1/0
  Crypto map tag: HQ-LAHORE-VPN, local addr 10.0.1.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
  current_peer 10.0.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 37, #pkts encrypt: 37, #pkts digest: 37
    #pkts decaps: 20, #pkts decrypt: 20, #pkts verify: 20
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

    local crypto endpt.: 10.0.1.2, remote crypto endpt.: 10.0.2.2
    path mtu 1500, ip mtu 1500, ip mtu idb Serial1/0
    current outbound spi: 0x60EE00E7(1626210535)

  inbound esp sas:
    spi: 0xAA081620(2852656672)
      transform: esp-aes esp-sha-hmac ,
      in use settings ={Tunnel, }
      conn id: 9, flow_id: SW:9, crypto map: HQ-LAHORE-VPN
      sa timing: remaining key lifetime (k/sec): (4434696/464)
      IV size: 16 bytes
      replay detection support: Y
      Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:
```

AAA AND SSH Verification

Administrative access was verified using the local admin account.

```
HQ-Router#ssh -l admin 192.168.20.1

Password:
AUTHORIZED ACCESS ONLY - IT-SIMPLERA SOLUTIONS
Lahore-Router>
Lahore-Router>en
```

Traceroute verification

```
PC1> trace 192.168.20.10
trace to 192.168.20.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    40.635 ms  31.116 ms  30.326 ms
 2      * * *
 3      * * *
 4      * * *
 5      * * *
 6      * * *
 7      * * *
 8      * * *

PC1> trace 192.168.30.10
trace to 192.168.30.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    39.346 ms  33.058 ms  28.043 ms
 2      * * *
 3      * * *
 4      * * *
 5      * * *
 6      * * *
 7      * * *
 8      * * *
```

Troubleshooting Report

Troubleshooting was an important part of the implementation.

Issue 1 — PC1 Could Not Reach HQ Gateway

Symptom

PC1 initially reported: host (192.168.10.1) not reachable

Investigation

The switch was inspected using:

```
show interfaces status
show interfaces gi0/1
show cdp neighbors
```

The router interface itself was operational, but the endpoint-to-switch connectivity required further investigation.

Resolution

The HQ switch and endpoint connectivity were corrected and verified.

After correction: PC1> ping 192.168.10.1

returned five successful replies.

Issue 2 — PC3 Could Not Reach Lahore Gateway

Symptom

PC3 initially reported: host (192.168.20.1) not reachable

Investigation

The Lahore router showed:

```
FastEthernet0/0
```

```
192.168.20.1
up
up
```

CDP confirmed that the switch was physically connected to the Lahore router.

However, the switch/endpoint environment was not forwarding PC traffic correctly.

Resolution

The original Lahore switch was removed and reattached with the three PCs. The VPCS nodes were subsequently executed within the GNS3 VM environment.

After rebuilding the local connectivity:

```
PC3> ping 192.168.20.1
```

returned successful replies.

This demonstrated that the problem was primarily related to the endpoint/switch/GNS3 environment rather than the Lahore router's Layer-3 interface.

VPN Not Establishing

Initially: `show crypto isakmp sa`

returned no active Security Association.

Similarly:

```
#pkts encaps: 0
#pkts encrypt: 0
#pkts decaps: 0
#pkts decrypt: 0
```

Root Cause

IPsec crypto maps do not necessarily establish a tunnel immediately after configuration. The configured interesting traffic must be generated to trigger negotiation.

Resolution

Traffic was generated between the HQ and Lahore LANs:

```
PC1> ping 192.168.20.10
```

After the traffic was generated, the tunnel became active.

The resulting state was: `QM_IDLE`

and IPsec packet counters began increasing.

GNS3 Switch / VPCS Connectivity

One of the major practical challenges was that the initial VPCS execution environment caused inconsistent LAN connectivity.

The investigation included:

```
show cdp neighbors
show interfaces status
show interfaces gi0/x
show spanning-tree
```

The replacement switch showed:

```
Gi0/0 connected  
Gi0/1 connected
```

and spanning tree showed the active ports in:

```
Desg FWD
```

After moving the VPCS execution to the GNS3 VM, PC1, PC2, and PC3 successfully reached their respective gateways.

GNS3 Configuration Persistence

During the project, a major operational problem was observed where reopening the GNS3 project caused previously configured crypto/security commands to appear missing.

For example, after reopening the project, Karachi-Router showed: `show crypto isakmp policy` with only the default protection suite and: `show crypto map`, No crypto maps found.

Root Cause

The issue was related to GNS3 device/project persistence rather than the correctness of the commands themselves.

Preventive Procedure

The following procedure should be followed after every major configuration stage: `write memory`

Then verify:

```
show startup-config  
show running-config
```

The GNS3 project should also be saved before closing.

For final submission, configuration exports should be stored separately in the GNS3 folder as an additional backup.

Challenges Faced

Several challenges were encountered during implementation.

1. GNS3 Resource and Appliance Behavior

The Cisco c3725 routers and GNS3 VM environment required careful resource management. Incorrect appliance behavior or excessive resource consumption could affect connectivity.

2. VPCS Connectivity

The initial VPCS/switch environment caused local gateway connectivity failures. Replacing the Lahore switch and running the VPCS nodes through the GNS3 VM resolved the issue.

3. VPN Tunnel Activation

The IPsec tunnel initially showed no Security Association because no interesting traffic had triggered the negotiation. Generating traffic between the protected LANs established the tunnel.

4. Crypto Configuration Persistence

Security configurations appeared absent after reopening the project. This highlighted the importance of saving the router startup configuration and maintaining external configuration backups.

This approach prevented VPN problems from being incorrectly diagnosed when the underlying LAN connectivity was not yet functional.

Security Assessment

The implemented architecture provides several security improvements over an unsecured enterprise network.

Threat	Implemented Control	Mitigation
WAN traffic interception	IPsec	Encrypts traffic
Unauthorized router login	AAA	Authenticated access
Credential interception	SSHv2	Encrypted management
Telnet attacks	SSH-only VTY	Removes insecure Telnet
Rogue OSPF updates	MD5 authentication	Protects routing adjacency
Unrestricted traffic	ZBF	Stateful filtering
Device reconnaissance	Hardening	Reduces exposed services
Event investigation difficulty	Syslog	Central event logging
Incorrect timestamps	NTP	Consistent time
Monitoring security	SNMPv3	Authenticated/secure monitoring

The architecture therefore implements **defense in depth**, where compromise or failure of one security control does not automatically expose the entire enterprise.

Conclusion

The Week 07 Enterprise VPN, Firewall, and Secure Access Implementation was successfully designed and implemented in GNS3.

The project established secure enterprise communication through IPsec Site-to-Site VPN, with IKE Phase 1 and IPsec Phase 2 successfully negotiated. The active QM_IDLE state, active IPsec Security Associations, and increasing encryption/decryption counters provided direct evidence of successful VPN operation.

AAA authentication and SSH Version 2 were implemented to secure administrative access, while OSPF authentication and passive interfaces strengthened the routing infrastructure. Zone-Based Firewall policies provided stateful traffic control, and Syslog, NTP, and SNMPv3 provided monitoring and operational visibility.

Overall, the completed implementation demonstrates a layered enterprise security architecture combining **confidentiality, authentication, traffic control, routing integrity, secure administration, and centralized monitoring** in a simulated production-style network.